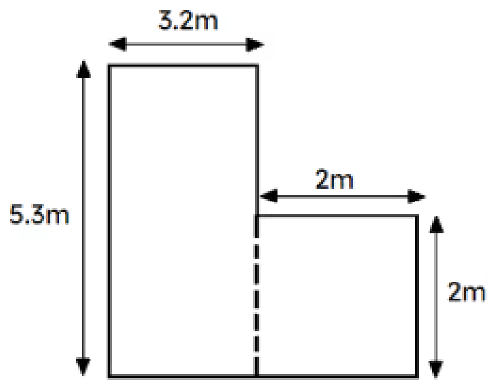
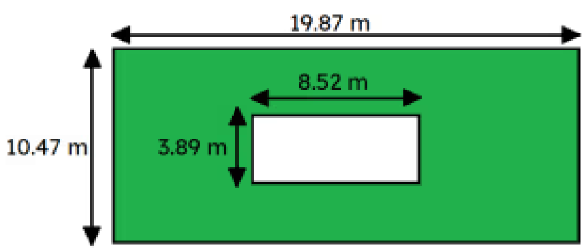
Problem solving with estimation and rounding

1 Work out an estimate for the area of the shape. Tick **1** correct answer



- [ ] 5 m^2
- [ ] 10 m^2
- [ ] 12 m^2
- [x] 19 m^2
- [ ] 25 m^2

2 Here is a grassed garden with a summer house in the centre. Using estimation and rounding, estimate the area of the grass. Tick **1** correct answer



- [ ] 50 m^2
- [ ] 90 m^2
- [ ] 100 m^2
- [x] 164 m^2
- [ ] 800 m^2

3 A piece of string was 32 cm measured to the nearest cm. A length of 5.6 cm (to 1 d.p) was cut off. What is the error interval for the remaining length x cm? Tick **1** correct answer

- [x] $25.85 < x < 26.95$
- [ ] $25.85 \leq x < 26.95$
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- [ ] $25.95 \leq x < 26.85$

4 Complete the statement. Fermi problems encourage: Tick **3** correct answers

- [x] critical thinking and well calculated estimates using data or information
- [x] openness of answers and flexibility so answers can be achieved in different ways
- [x] an appreciation there are a range of answers, the focus is on the working out
- [ ] random guesses but use estimation and or rounding

5 Fermi problem : How long will it take to count to 1 million (without breaks, sleep, etc)? Tick **1** correct answer

- [ ] 50 days
- [ ] 40 days
- [x] 16 days
- [ ] 5 days
- [ ] 3 days

6 Fermi problem : If you put 1 million pound coins in a box, what is the volume of the box in m^3 ? Tick **1** correct answer

- [ ] 0.5 m^3
- [x] 1.2 m^3
- [ ] 5 m^3
- [ ] 8 m^3
- [ ] 12 m^3